

Question				Marking details				Marks available													
								AO1	AO2	AO3	Total	Maths	Prac								
4	(a)			<table><tr><td>Radio waves</td><td>Microwaves</td><td>Infra-red</td><td>Visible</td></tr><tr><td>$4 \times 10^2 \text{ m}$</td><td>$2 \times 10^{-2} \text{ m}$</td><td>$3 \times 10^{-5} \text{ m}$</td><td>$5 \times 10^{-7} \text{ m}$</td></tr></table>				Radio waves	Microwaves	Infra-red	Visible	$4 \times 10^2 \text{ m}$	$2 \times 10^{-2} \text{ m}$	$3 \times 10^{-5} \text{ m}$	$5 \times 10^{-7} \text{ m}$		2		2		
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All correct award 2 marks 2 or 3 correct award 1 mark 1 correct 0 marks																					
	(b)	(i)		UV, X- <u>rays</u> , gamma [rays] – all 3 needed				1			1										
		(ii)		[Radiation that] has high energy / interacts with atoms / removes electrons from atoms / damages <u>cells</u> Don't accept causes cancer / mutations				1			1										
	(c)			Transfer energy / travel at the same speed [in a vacuum] / can travel through empty space Don't accept transmit information				1			1										